Mahbub Ali

Software Developer | Aspiring AI Researcher

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Profile

Computer Science and Engineering graduate of Green University of Bangladesh, with a CGPA of 3.54 and over three years of professional software development. The three years of practice were on HealthConnect, PrimeMIS, and Muallim App. The current direction is a Master’s by Research in Computer Science. This curriculum vitae is prepared for your university and for the supervisors who review it. The research focus is healthcare artificial intelligence, machine learning, predictive analytics, and early disease risk detection. The career goal is to become an AI researcher specializing in healthcare AI.

Education

Bachelor’s in Computer Science and Engineering, Green University of Bangladesh. CGPA 3.54.

A Medium of Instruction certificate is available and can be shared with admissions offices and prospective supervisors on request.

Intended postgraduate study, not a current enrolment: Master’s by Research in Computer Science, prepared for your university. Research focus: Healthcare AI, Machine Learning, Predictive Analytics, and Early Disease Risk Detection.

Professional experience

Software Developer, 3+ years. The work in that period was HealthConnect, PrimeMIS, and Muallim App.

HealthConnect. React Native and Expo healthcare application, published on Google Play and the App Store.

- Doctor search and appointment booking.
- Patient reports and digital prescriptions.
- Home doctor, nurse, and physiotherapy services.
- Firebase integration, REST APIs, authentication, and notifications.

Technologies: React Native, Expo, Firebase, REST APIs.

PrimeMIS. Hospital management information system with an Angular frontend, a .NET Web API backend, and MySQL.

- Patient and doctor management.
- Appointment management.
- Pharmacy and laboratory management.
- Reports and digital prescription workflows.

Technologies: Angular, .NET Web API, MySQL.

Muallim App. React Native mobile application, separate from the healthcare systems above. Published on the App Store.

SakuraEHealth

SakuraEHealth is the niche company being established for health-related software. The products below are separate systems. They are not one deployed platform, and they are not the three-year employment record above.

MediScan Pro. Low-cost offline endoscopy platform for Android phones and tablets, connected over USB. The capture path runs from patient intake to a local HTML report. An Android module scores otoscopy images on device and marks a low-confidence result as uncertain. That score is a suggestion, not a diagnosis.

- Field summary, Dhaka, May 2025: 15 doctors used the platform in 120 endoscopy procedures. A report took about 10-15 minutes, compared with 2-3 days for the usual diagnosis path. Ease of use was rated 7 out of 10. The reported investment could be recovered after 15-20 procedures.
- That summary is a workflow result. It does not report sensitivity or specificity for the on-device classifier.

Technologies: React Native, Expo, Kotlin, TensorFlow, TensorFlow Lite.

Sakura ENT Center. Healthcare website and appointment system, with patient queue and sequence display through Sakura TV. Public site: sakuraentcenter.com.

Technologies: Angular, NestJS, PostgreSQL, Socket.IO, Kotlin.

Prescription writing. Next.js workspace for a medicine catalogue, interaction records, specialty templates, and draft or final prescriptions, with a link to the Sakura appointment API.

Technologies: Next.js, TypeScript, PostgreSQL, Drizzle, PowerSync.

Hospital Management. The deployed hospital system is PrimeMIS, listed under professional experience.

EHealth App. Portfolio name for the application published as HealthConnect, listed under professional experience. The case study treats booking consistency, separate home-visit workflows, clinical documents on a phone, and authentication.

SakuraEHealth website. Public company site. The Angular source was first built as a studio site and is presented under the company name. Marketing figures from that source are not repeated.

Technical skills

Frontend: React, React Native, Expo, Angular, Next.js, TypeScript, Tailwind CSS.

Backend: Node.js, Express.js, NestJS, .NET Web API.

Database: MongoDB, MySQL, PostgreSQL.

Tools: Git, GitHub, Firebase, Azure DevOps.

Evidenced in project source: TensorFlow, TensorFlow Lite, and Kotlin, from MediScan Pro and Sakura TV. This is implementation evidence, not a published classifier accuracy figure.

Currently learning, for research preparation: Python, Pandas, NumPy, Scikit-learn, and Machine Learning. These are not presented as completed research practice.

Research preparation

- Drafting a proposal on early disease risk detection from electronic health records.
- Looking for a public or approved dataset. No dataset has been selected, and no analysis has been completed.

- Studying logistic regression, random forests, gradient boosting, and support vector machines.
- Studying Python, Pandas, and NumPy for tabular health data.
- Studying SHAP. It has not been applied to a project model.

Proposed research

Working title: Machine Learning-Based Predictive Analytics for Early Disease Risk Detection Using Electronic Health Records.

Status: proposal in preparation for a Master’s by Research in Computer Science. It is not a publication, and it does not report trained models or scores. The full proposal is the companion PDF.